RAJALAKSHMI ENGINEERING COLLEGE

Ahmed Waseem 250701028-2025-CSE

A2

Started on	Wednesday, 5 August 2026, 8:40 AM
State	Finished
Completed on	Wednesday, 5 August 2026, 9:12 AM
Time taken	32 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int i = 1;
    int s = 1;
    int count = 2;   // i=1 and s=1

    while (1)
    {
        count++;      // while condition
        if (!(s <= n))
            break;

        i++;
        count++;      // i++

        s += i;
        count++;      // s += i
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Ahmed Waseem 250701028-2025-CSE

A2

Started on	Wednesday, 5 August 2026, 8:50 AM
State	Finished
Completed on	Wednesday, 5 August 2026, 9:12 AM
Time taken	21 mins 44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int n, count = 0;
    scanf("%d", &n);

    count++;

    if(n == 1)
    {
        count++;
    }
    else
    {
        for(int i = 1; ; i++)
        {
            count++;

            if(i > n)
                break;

            for(int j = 1; ; j++)
            {
                count++;

                if(j > n)
                    break;

                count++;
                count++;
                count++;
                break;
            }
        }
```

```
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Ahmed Waseem 250701028-2025-CSE

A2

Started on	Wednesday, 5 August 2026, 8:56 AM
State	Finished
Completed on	Wednesday, 5 August 2026, 9:12 AM
Time taken	16 mins 1 sec
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int num, count = 0;
    scanf("%d", &num);

    for (int i = 1; ; i++)
    {
        count++;            // for loop condition

        if (i > num)
            break;

        count++;            // if condition (num % i == 0)

        if (num % i == 0)
        {
            count++;        // printf("%d ", i)
        }
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Ahmed Waseem 250701028-2025-CSE

A2

Started on	Wednesday, 5 August 2026, 8:58 AM
State	Finished
Completed on	Wednesday, 5 August 2026, 9:12 AM
Time taken	13 mins 42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int count = 1;   // Count: c = 0

    for (int i = n / 2; i < n; i++)
    {
        count++;   // i < n

        for (int j = 1; j < n; j = j * 2)
        {
            count++;   // j < n

            for (int k = 1; k < n; k = k * 2)
            {
                count++;   // k < n
                count++;   // c++
            }
            count++;       // last failed k < n
        }
        count++;           // last failed j < n
    }
    count++;               // last failed i < n

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Ahmed Waseem 250701028-2025-CSE

A2

Started on	Wednesday, 5 August 2026, 9:07 AM
State	Finished
Completed on	Wednesday, 5 August 2026, 9:11 AM
Time taken	4 mins 35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d",&n);

    int rev = 0, remainder;
    int count = 2;

    while(n != 0)
    {
        count++;
        remainder = n % 10;
        count++;

        rev = rev * 10 + remainder;
        count++;

        n = n / 10;
        count++;
    }

    count++;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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